

APD background fits: processing and sphere residuals

Draft for discussion with Richard · 6 October 2026

1. Data processing: uniform final-arc references

Aim. Compare physical models against the same ordered two-loop trajectory, with roughly equal weight per unit anisotropy arc. The measured reference is constructed before any background fit. No fitted cone or direct-inversion sphere circle determines its shape.

Calibration and optics. APD voltages are reordered from (90, 45, 135, 0) to (0, 90, 45, 135). Historical channel gains precede the current inverse transmission matrix, which is unchanged. The forward signal uses $\text{NA}_{\text{in}} = 0.38$, $\text{NA}_{\text{out}} = 1.3$, water/oil indices 1.33/1.51, Fresnel transmission and corrected BFP mapping.

Cycles and averaging. Retain the existing complete-cycle detection and ordered 128-bin channel representation. Alternate accepted cycles form training and held-out groups: 152/152 for 30–31 s and 109/108 for 10–11 s. Average corresponding *four-channel intensities* across cycles in each group. The resulting final-average anisotropy points are not generally equally spaced, even if each cycle was binned by equal guide arc.

Redistribute the final training average. Interpolate linearly between consecutive four-channel mean vectors, closing last to first. Let $\mathbf{I}(t)$ be this interpolant, with original-bin coordinate $0 \leq t \leq 128$. Calculate

$$X(t) = \frac{I_0(t) - I_{90}(t)}{I_0(t) + I_{90}(t)}, \quad Y(t) = \frac{I_{45}(t) - I_{135}(t)}{I_{45}(t) + I_{135}(t)}.$$

Integrate cumulative distance along the *ordered* path, counting both loops and every repeated pass:

$$s(t) = \frac{\int_0^t \sqrt{X'(v)^2 + Y'(v)^2} dv}{\int_0^{128} \sqrt{X'(v)^2 + Y'(v)^2} dv}, \quad t_j = s^{-1}(j/128), \quad j = 0, \dots, 127.$$

Numerically use 128 subdivisions per original segment. The new training reference is $\mathbf{I}(t_j)$, retaining all four intensities. Apply the *same* original-bin interpolation indices and weights to the held-out average. Held-out points are not independently made uniform or realigned. Equal arc does not require equal chords around sharp bends. Resampling correlates adjacent points and adds no independent information.

Fit with frozen observations. Simulate a dense mechanical revolution for a proposed cone, brightness and background. Compute its predicted *measured* intensities, including interference, then its own anisotropy arc. Sample those intensities at

$$s_j^{\text{model}} = (\delta + \epsilon u_j) \bmod 1, \quad \epsilon = \pm 1.$$

Here u_j are the actual arc coordinates of the final resampled training reference (recomputed to remove small discretization differences). One cyclic offset δ is fitted per window and both directions are tested. There is no nearest-point matching or fitted phase warp.

$$J = \sum_j \sum_{i=1}^4 \left[\frac{I_{i,j}^{\text{pred}} - I_{i,j}^{\text{train}}}{T_j^{\text{train}}} \right]^2, \quad T_j = \sum_i I_{i,j}.$$

Uniform bins remove the main anisotropy-density weighting; the normalization still weights relative intensity error. This is not uniform mechanical phase or a noise-calibrated likelihood. Held-out cycles assess repeatability with frozen parameters, not independent absolute-angle accuracy. The following diagnostics reuse the existing fits; no sphere residual was optimized.

2. Signal, background models and orientation diagnostics

For rod direction $\mathbf{d} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$ and analyser angle ψ_i ,

$$Q_i(\mathbf{d}) = A_F + B_F \sin^2 \theta + C_F \sin^2 \theta \cos 2(\phi - \psi_i),$$

$$S_i(\mathbf{d}) = a_0^2 \sin^2 \theta Q_i(\mathbf{d}), \quad T_S = 4a_0^2 \sin^2 \theta (A_F + B_F \sin^2 \theta).$$

One constant a_0 per window; the $\sin^2 \theta$ factor is circular-excitation intensity, not an independently fitted per-point brightness. $A_F = 0.890806$, $B_F = 0.109194$, $C_F = 0.927781$ in the adopted normalization.

2A. Background only (8 parameters/window).

$$I_i = S_i + B_i, \quad \mathbf{B} = \frac{B_{\text{tot}}}{4} (1 + f_x, 1 - f_x, 1 + f_y, 1 - f_y), \quad |f_x|, |f_y| \leq 1.$$

Three cone parameters, a_0 , cyclic offset, and three background parameters. Equal background pair sums, nonnegative channels, no Stokes-disk restriction. $T = T_S + B_{\text{tot}}$, with constant $B_{\text{tot}} \leq 0.95 \min T_{\text{train,original}}$. Sphere diagnostic: $\hat{S}_i = I_i^{\text{heldout}} - B_i$.

2B. Richard exact four-C (9 parameters/window).

$$I_i = S_i + C_i \sqrt{S_i}, \quad T = T_S + \sum_i C_i \sqrt{S_i}.$$

Three cone parameters, a_0 , cyclic offset and four constant C_i . There is no additive background power in this approximation. Its correction varies with orientation through S_i . Numerical parameters are $\alpha = a_0^2$, $K_i = a_0 C_i$; selected $\alpha > 0$, but the zero-brightness/divergent- C limit remains structurally possible. For the sphere diagnostic, invert each *measured* channel exactly:

$$\hat{S}_i = \left[\frac{-C_i + \sqrt{C_i^2 + 4I_i^{\text{heldout}}}}{2} \right]^2.$$

Measured channels are positive in these references, giving one positive amplitude root. Do not subtract a cross term evaluated at the predicted signal for this model.

2C. General stationary field (33 parameters jointly).

$$E_{s,i}(\mathbf{d}) = a_0(d_x + i d_y) \mathcal{L}_i \mathbf{d},$$

$$I_i = S_i + L_i(\mathbf{d}) + B_i, \quad L_i = 2 \operatorname{Re} \langle E_{s,i}, E_{b,i} \rangle, \quad B_i = \|E_{b,i}\|^2 + N_i,$$

$$T = T_S + \sum_i L_i(\mathbf{d}) + \sum_i (\|E_{b,i}\|^2 + N_i).$$

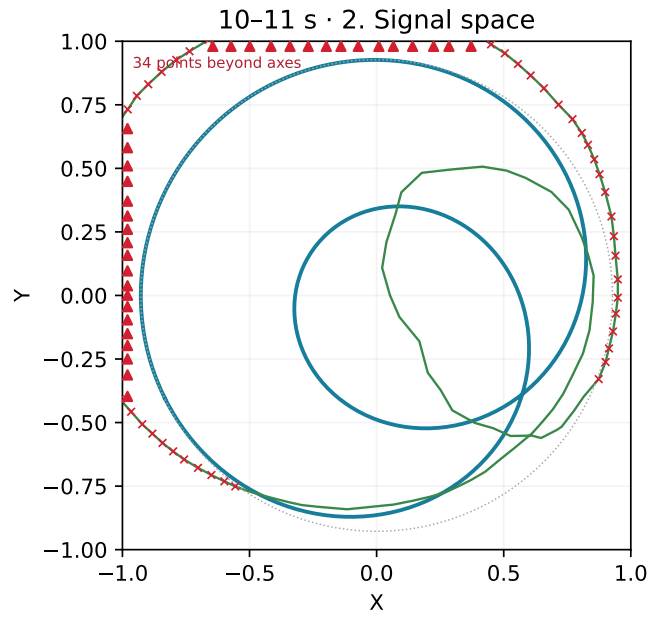
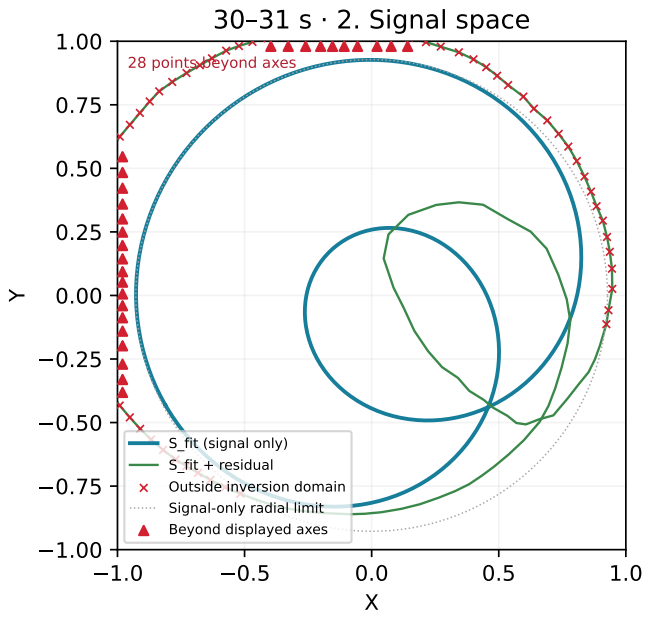
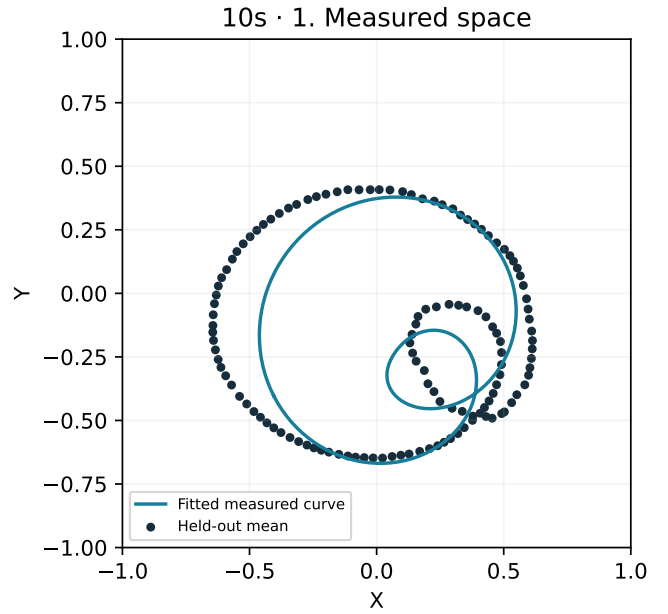
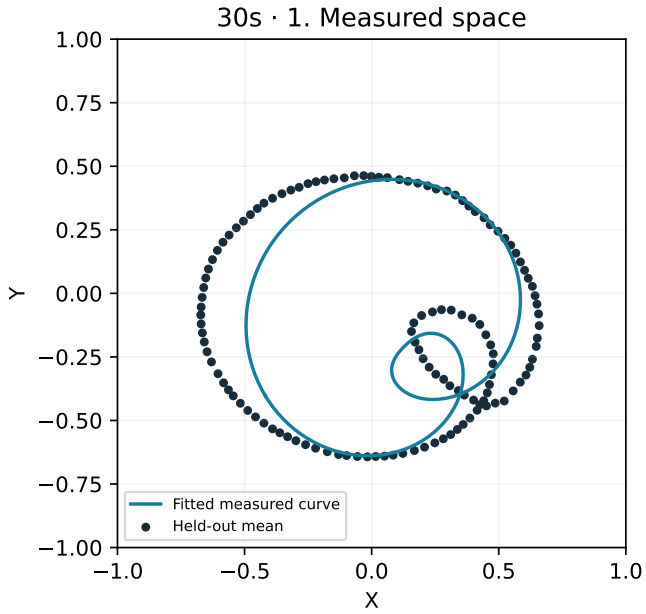
Ten complex transverse pupil modes describe one fixed common field shared by both windows; N_i is a constant non-interfering positive-Stokes background. There are 23 shared background parameters and five cone/brightness/offset parameters per window. B_i stays constant; L_i changes through the predicted rod field, not free per-point coefficients. The shared total background is capped at 95% of the stricter original minimum intensity. Generality assumes fixed-position ideal-dipole common-pupil optics.

Sphere diagnostic uses $\hat{S}_{i,j} = I_{i,j}^{\text{heldout}} - B_i - L_i(\mathbf{d}_j^{\text{fit}})$. This is a *conditional subtraction at the matched fitted orientation*, not a self-consistent inverse of the full field model. It shows how its intensity residual is amplified by signal-only inversion. The three panels therefore compare explicitly stated correction diagnostics, not identical inverse estimators.

2A. Background only

Four-channel fits and corrected held-out points; no new fit to sphere coordinates.

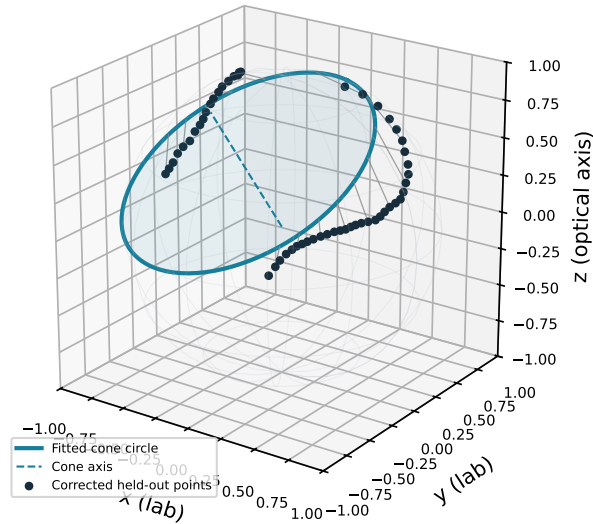
Background only · fixed correspondence, held-out diagnostics



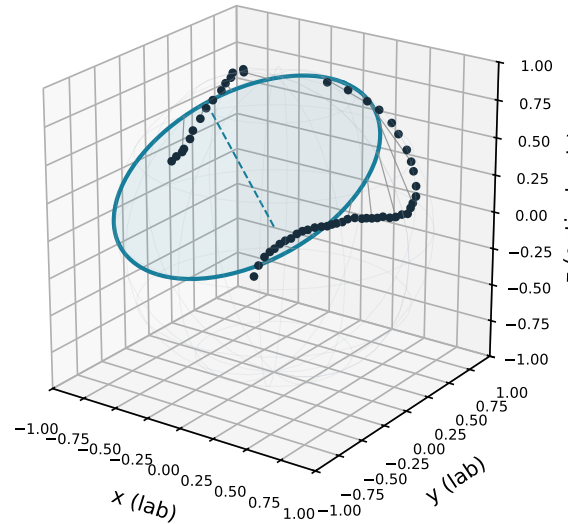
Sphere and cone-frame residuals

Background only · sphere and cone-frame residuals

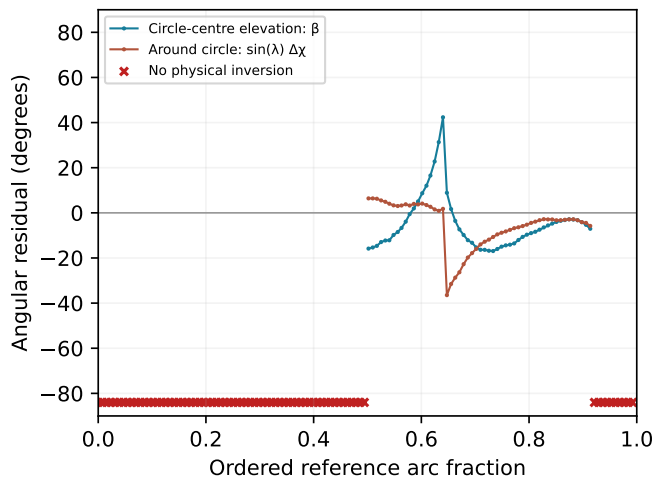
Unit sphere · 54/128 points invertible



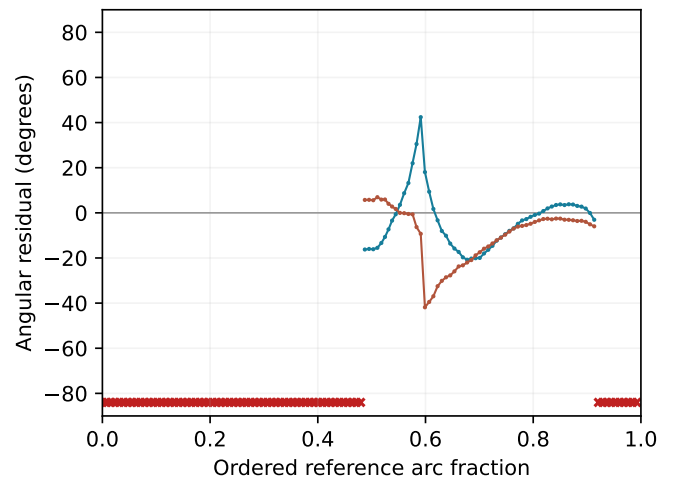
Unit sphere · 56/128 points invertible



Residuals in fitted-cone frame



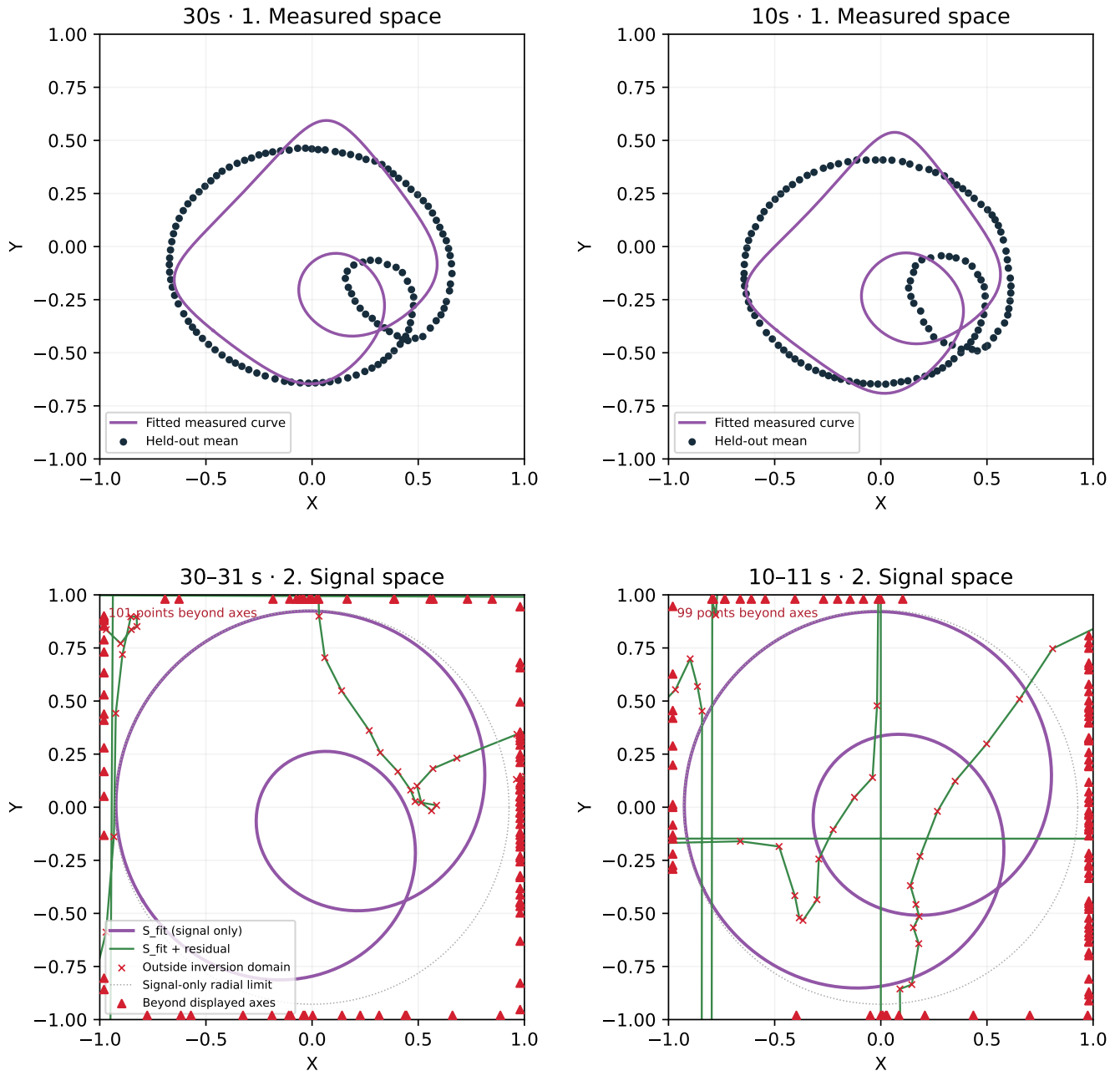
Residuals in fitted-cone frame



2B. Richard exact four-C

Sphere points use the exact channelwise quadratic inverse with the fitted constants.

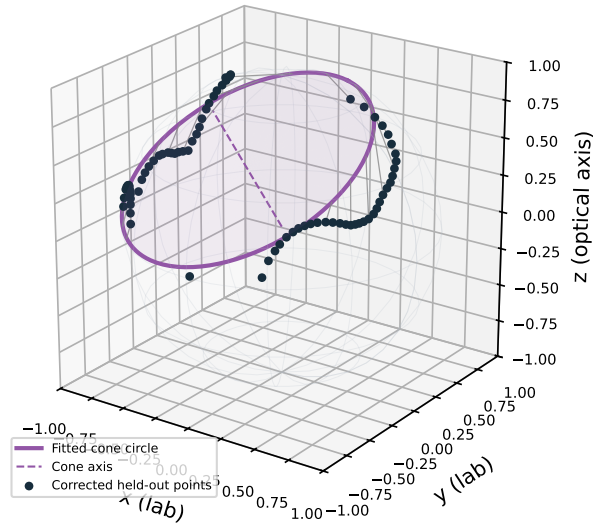
Richard four-C · fixed correspondence, held-out diagnostics



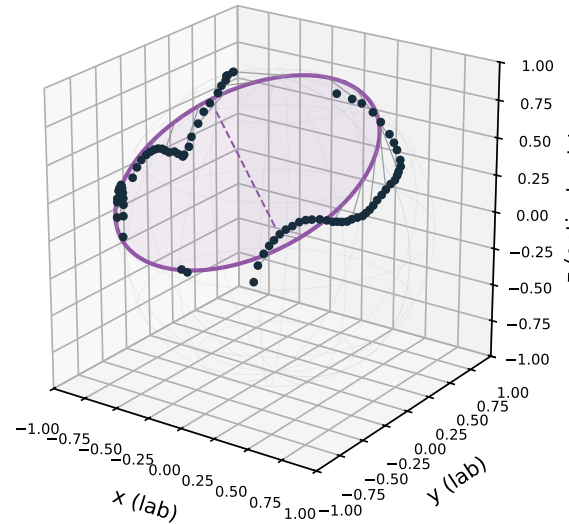
Sphere and cone-frame residuals

Richard four-C · sphere and cone-frame residuals

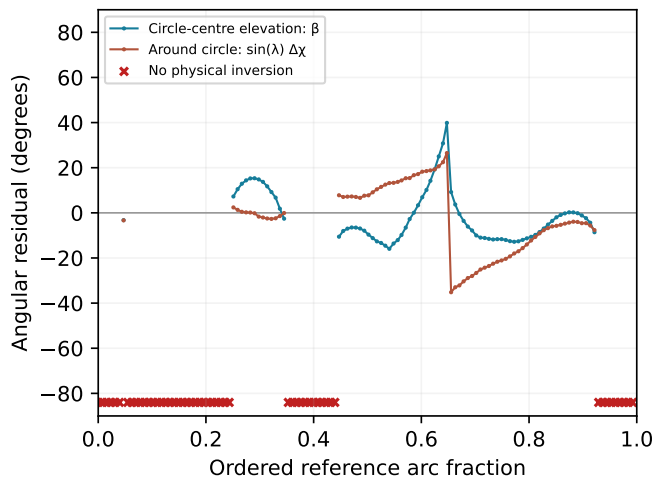
Unit sphere · 76/128 points invertible



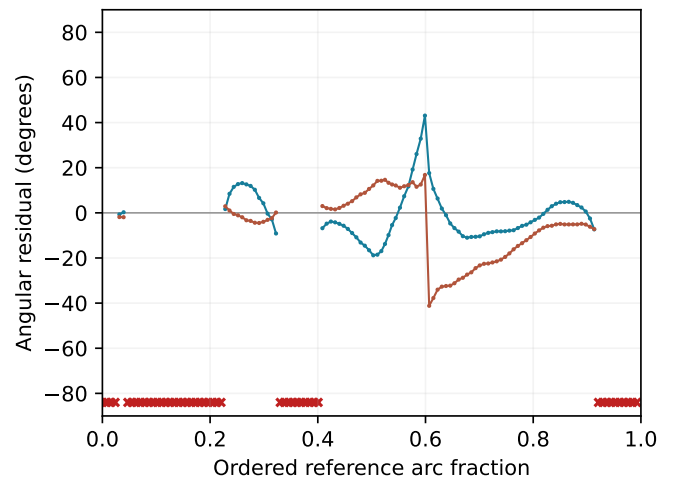
Unit sphere · 81/128 points invertible



Residuals in fitted-cone frame



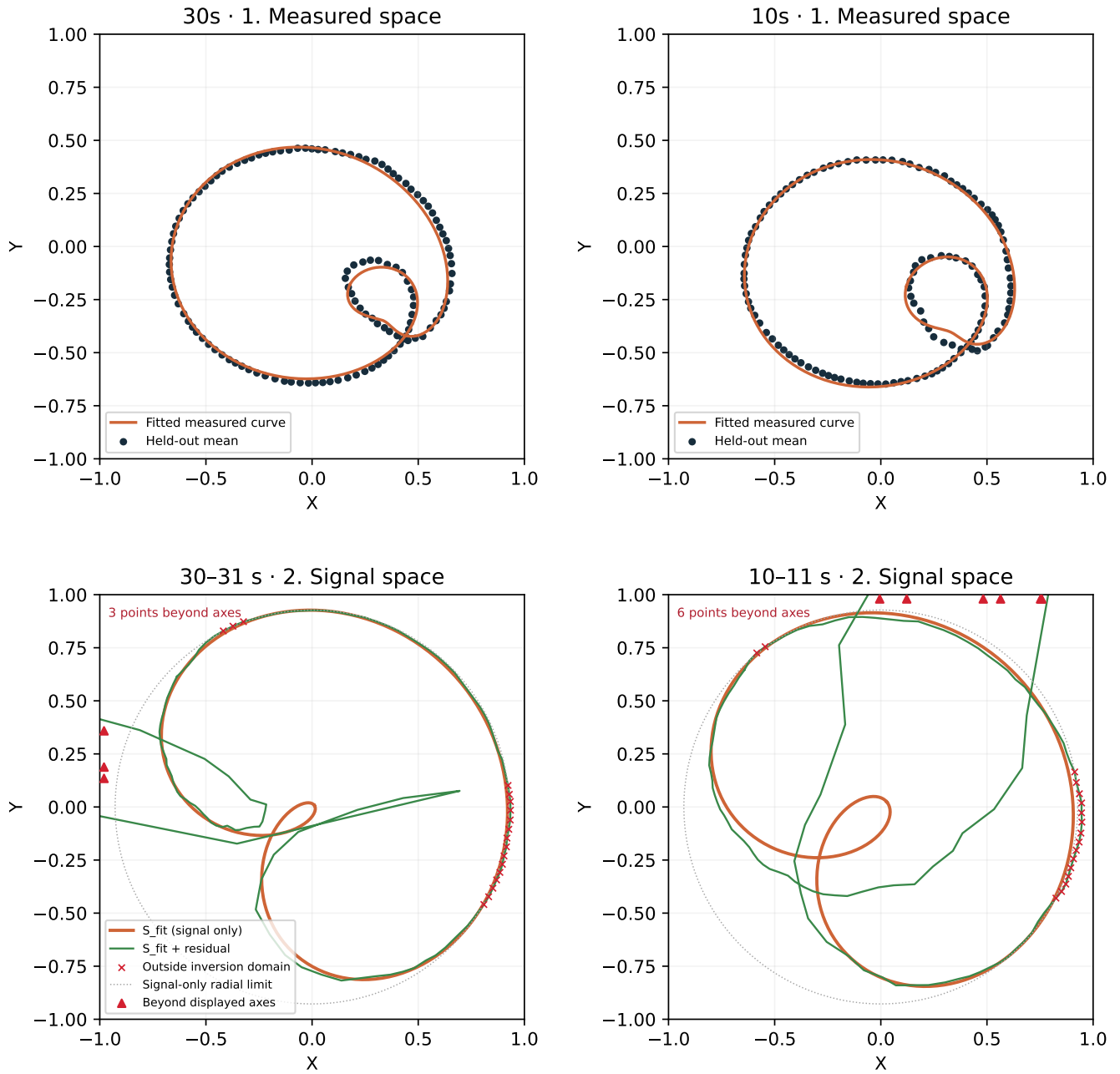
Residuals in fitted-cone frame



2C. General stationary field

Shared field across both windows. Sphere points use conditional fitted cross-term subtraction.

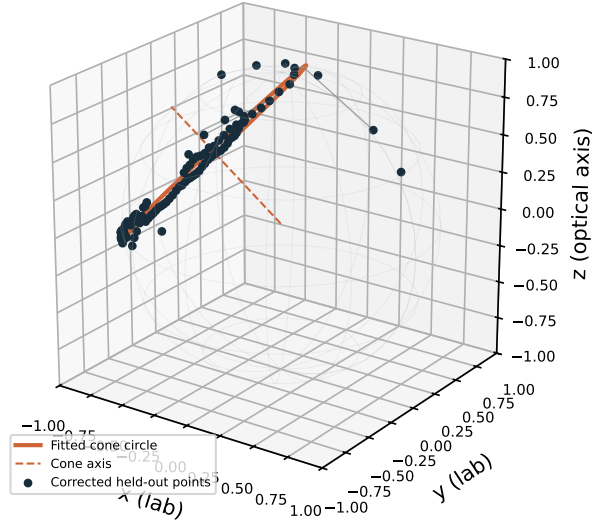
General stationary field · fixed correspondence, held-out diagnostics



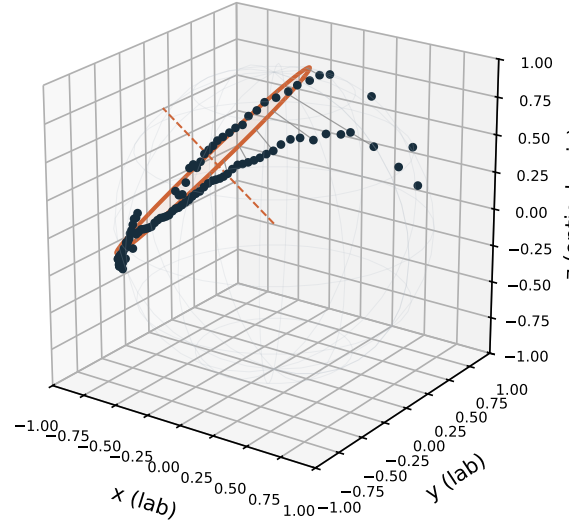
Sphere and cone-frame residuals

General stationary field · sphere and cone-frame residuals

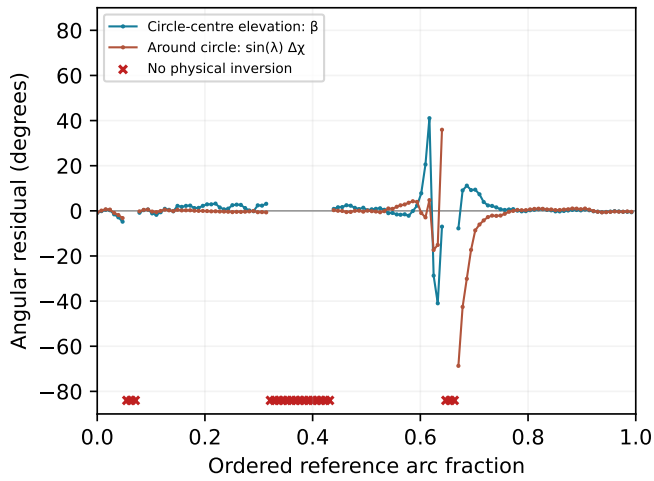
Unit sphere · 107/128 points invertible



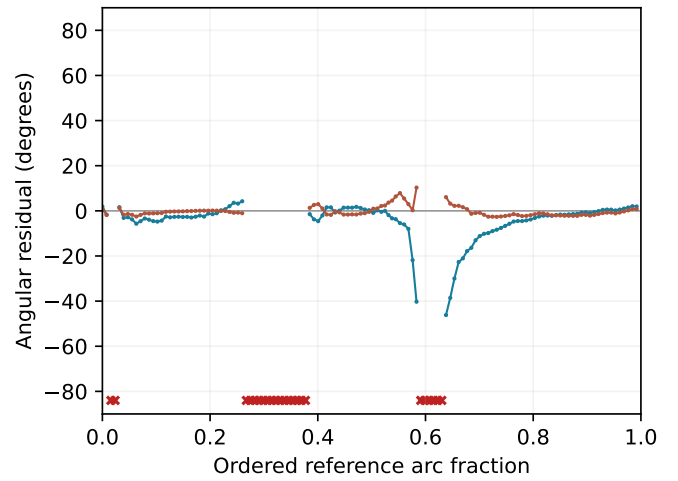
Unit sphere · 105/128 points invertible



Residuals in fitted-cone frame



Residuals in fitted-cone frame



Reading the unit-sphere and cone-frame residual plots

From corrected channels, calculate $(\hat{X}, \hat{Y})$, $r = \sqrt{\hat{X}^2 + \hat{Y}^2}$, then

$$\sin^2 \hat{\theta} = \frac{A_F r}{C_F - B_F r}, \quad \hat{\phi} = \frac{1}{2} \text{atan2}(\hat{Y}, \hat{X}).$$

Only nonnegative corrected channels and $0 \leq \sin^2 \hat{\theta} \leq 1$ are plotted. Points outside this domain are marked with red crosses along the bottom residual panel, not clipped onto the equator. This necessary domain test does not enforce pair-sum balance or the full brightness law.

Anisotropy overlays. Each model has separate measured-space (held-out mean versus fitted measured curve) and signal-space plots ($S^{\text{fit}} + \varepsilon$ versus S^{fit}). Green is calculated from $S^{\text{fit}} + \varepsilon$, with $\varepsilon = I^{\text{heldout}} - I^{\text{pred}}$, for all three models. Red crosses flag negative corrected channels or an anisotropy outside the signal-only inversion domain. These coordinates are not physically clipped. Triangles at the plot boundary indicate points beyond the fixed $[-1, 1]$ axes; they are display indicators, not reconstructed positions. The dotted circle marks r_{max} . For Richard, this green frozen-cross-term diagnostic differs from the exact quadratic inverse used in its sphere panel; do not equate the two sets of points.

Branch choice. For each point, enumerate the four combinations $\hat{\phi}$ or $\hat{\phi} + \pi$, and $\hat{\theta}$ or $\pi - \hat{\theta}$. Choose the unit vector maximizing its dot product with the *already matched* fitted cone direction. This changes only a measurement-degenerate branch; it does not move the point onto the cone or choose a new correspondence. It is a cone-conditioned convention, not independent recovery of direction or proof of continuous branch tracking.

Laboratory sphere plots. All sphere coordinates retain the laboratory frame, with optical axis z . The fitted cone tilt is preserved; a dashed line shows its axis $\mathbf{k}$. The circle centre is $\mathbf{c} = \cos \lambda \mathbf{k}$, and the faint disk lies in its plane. Grey connectors show every fourth valid matched residual.

Cone-frame residuals. Rotate only for calculating residual coordinates, so z' follows $\mathbf{k}$. Draw the vector from the fitted circle centre to the corrected point, $\mathbf{v}_j = \hat{\mathbf{d}}_j - \mathbf{c}$. Its signed elevation above the circle plane is

$$\beta_j = \text{atan2}(\mathbf{v}_j \cdot \mathbf{k}, \|\mathbf{v}_j - (\mathbf{v}_j \cdot \mathbf{k})\mathbf{k}\|) = \text{atan2}\left(\hat{d}_{z,j}' - \cos \lambda, \sqrt{\hat{d}_{x,j}^2 + \hat{d}_{y,j}^2}\right).$$

Blue is β : zero on the fitted circle, positive on the cone-axis side of its plane. It is an angle measured from the *circle centre*, not lab θ or a polar-angle residual measured from the sphere centre.

$$\Delta\chi_j = \text{wrap}_{[-\pi, \pi)} \left[\text{atan2}(\hat{d}_{y,j}', \hat{d}_{x,j}') - \text{atan2}(d_{y,j}^{\text{fit}}, d_{x,j}^{\text{fit}}) \right].$$

Orange is $\sin \lambda \Delta\chi$, the signed displacement around the fitted circle on a unit sphere. It retains the original matched correspondence. Horizontal position is ordered reference arc, not time or uniformly spaced mechanical phase. For small departures, $\beta \simeq -(\hat{\alpha} - \lambda)$, where $\hat{\alpha}$ is polar angle from the cone axis, and squared geodesic error is approximately $\beta^2 + \sin^2 \lambda (\Delta\chi)^2$. For large deviations these approximations do not hold.

Model	Window	Invalid/128	Elevation RMS	Along RMS	Geodesic RMS
Background only	30s	74	12.9°	11.3°	17.1°
Background only	10s	72	13.1°	15.5°	20.4°
Richard	30s	52	11.5°	15.5°	19.0°
Richard	10s	47	11.0°	15.8°	18.9°
Stationary	30s	21	7.1°	9.6°	12.1°
Stationary	10s	23	9.6°	2.3°	11.4°

Interpretation. RMS values above include valid points only, with different valid subsets; they must not be used alone to rank models. Even the stationary fit leaves 21/23 points outside the signal-only inversion domain. Its close measured-anisotropy fit does not imply small or everywhere-defined corrected-angle errors. Near the equator the inversion is poorly conditioned; near low signal, subtraction magnifies residuals. Pair-sum mismatch is also retained rather than silently repaired.

Fit composition and limits of this draft

The following ranges describe the *forward fitted* signal and cross term at the 128 matched positions, normalized by original training maximum total intensity. They are not recovered experimental component measurements.

Model	Window	BG/ $T_{\max}$	Signal/ $T_{\max}$	Cross/ $T_{\max}$	Fitted folded θ
BG only	30s	35.5%	12.9–59.5%	0.0–0.0%	29.1–89.9°
BG only	10s	38.3%	15.5–54.5%	0.0–0.0%	33.7–89.9°
Richard	30s	—	1.7–7.7%	43.8–89.6%	29.0–88.6°
Richard	10s	—	0.6–2.0%	51.2–94.1%	33.4–88.4°
Stationary	30s	54.8%	1.3–110.8%	-80.5–5.1%	6.5–89.9°
Stationary	10s	54.5%	3.2–101.2%	-73.7–6.6%	10.9–89.9°

Background-only does not identify a physical coherent/noncoherent composition: it includes no interference term. Richard’s model specifies a signed correction, not an additive background power, so no physical BG percentage is inferred from it. For the selected stationary field, essentially 100% of represented background power is assigned to the coherent component, with negligible fitted N_i . This is conditional on this model and optimum; it is not a source measurement.

Model	Window	Held-out channel RMS	Corrected pair-imbalance RMS
Background only	30s	0.02871	0.70%
Background only	10s	0.03245	0.65%
Richard	30s	0.03908	10.14%
Richard	10s	0.03930	11.67%
Stationary	30s	0.00506	4.51%
Stationary	10s	0.00954	4.32%

Channel RMS is $\sqrt{\text{mean}_{i,j}[(I_{i,j}^{\text{pred}} - I_{i,j}^{\text{heldout}})/T_j^{\text{heldout}}]^2}$. Pair imbalance is $(\hat{S}_0 + \hat{S}_{90} - \hat{S}_{45} - \hat{S}_{135})/\sum_i \hat{S}_i$, using all bins, including those not invertible. A unit-sphere plot alone discards this additional inconsistency.

What the new plots establish. The stationary field gives much better forward agreement under fixed ordered correspondence, but residuals can become substantial or nonphysical after correction. The unit-sphere test exposes that amplification; it does not independently validate the fitted cone. Reference-conditioned branch selection chooses the visually closest permitted branch and therefore gives an optimistic branch-resolved comparison.

What has not changed. No model was refitted for this draft. The selected stationary optimization converged, but multiple starts do not certify a global optimum. Background is near its cap: 54.84% of the smaller original maximum versus an allowed 55.69%. Another local solution around 47% background has narrower recovered angles and slightly poorer fit. Absolute angles and physical background composition remain unvalidated.

Numerical consistency. The stationary forward calculation uses cached pupil quadrature; sphere inversion uses the analytic annular coefficients above. The tested unit-signal difference is below 7.7×10^{-5} , much smaller than current residuals. Doubling forward curve density changes predicted channels by less than $5 \times 10^{-8} T_{\max}$. Invalid samples are retained in diagnostics, not hidden by clipping or selection. The sphere residuals are conditional diagnostics, not uncertainty estimates or a new fit objective.

Scripts and arrays: draft/build.py (sphere/branch/residual calculations), draft/document.py (this LaTeX report), draft/sphere_metrics.json and six sphere NPZ files. Original fits and decompositions remain in the parent directory. No commit or push.